

Vectors

Outline

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Definition and Transpose

Introduction: Definition and Transpose

- ▶ In this section we define what a **vector** is and fix the notation for its entries
- ▶ We then introduce the **transpose**, which turns a column vector into a row vector
- ▶ Vectors are the basic objects of linear algebra: they will be used to represent data, variables and solutions of optimization problems



Definition 1: Vector

A *vector* is a column of real numbers with n rows.

Given a vector $\mathbf{a} \in \mathbb{R}^n$, we write:

$$\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}.$$

Terminology:

- ▶ The integer n is the **dimension** of the vector (the number of rows).
- ▶ The quantity $[\mathbf{a}]_i$ (or a_i) denotes the **i -th entry** (or **i -th component**) of $\mathbf{a}$, for $i \in \{1, 2, \dots, n\}$.

Example

The vector

$$\mathbf{a} = \begin{pmatrix} 2 \\ -5 \\ 3 \\ 1 \end{pmatrix} \in \mathbb{R}^4$$

is a column vector with $n = 4$ rows, where $a_1 = 2$, $a_2 = -5$, $a_3 = 3$, and $a_4 = 1$.

Definition 2: Transpose of a vector

Given a column vector $\mathbf{a} \in \mathbb{R}^n$, the **transpose** of $\mathbf{a}$, denoted by $\mathbf{a}'$, is the row vector:

$$\mathbf{a}' = (a_1 \quad a_2 \quad \dots \quad a_n) \in \mathbb{R}^{1 \times n}. \quad (1)$$

Example

Given the column vector

$$\mathbf{a} = \begin{pmatrix} 2 \\ -5 \\ 3 \\ 1 \end{pmatrix} \in \mathbb{R}^4,$$

its transpose is the row vector

$$\mathbf{a}' = (2 \quad -5 \quad 3 \quad 1) \in \mathbb{R}^{1 \times 4}.$$

Summary

Vectors:

- ▶ A vector $\mathbf{a} \in \mathbb{R}^n$ is a column of n real numbers
- ▶ n is its dimension, a_i (or $[\mathbf{a}]_i$) is its i -th entry

Transpose:

- ▶ $\mathbf{a}' = (a_1 \ a_2 \ \dots \ a_n) \in \mathbb{R}^{1 \times n}$ is the row vector with the same entries

Sum and Scalar Multiplication



Introduction: Sum and Scalar Multiplication

- ▶ We now define two fundamental operations on vectors: addition and scalar multiplication
- ▶ Both operations act **component by component**
- ▶ Together with their properties, they are the building blocks of linear combinations, which we will study at the end of this chapter

Sum of vectors

Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n$$

their **sum** is the vector

$$\mathbf{p} + \mathbf{w} = \begin{pmatrix} p_1 + w_1 \\ p_2 + w_2 \\ \vdots \\ p_n + w_n \end{pmatrix} \in \mathbb{R}^n \quad (2)$$

Scalar multiplication

Given a scalar $\lambda \in \mathbb{R}$ and a column vector

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n$$

the **scalar multiplication** of $\mathbf{p}$ by λ is the vector

$$\lambda \mathbf{p} = \begin{pmatrix} \lambda p_1 \\ \lambda p_2 \\ \vdots \\ \lambda p_n \end{pmatrix} \in \mathbb{R}^n \quad (3)$$

Example

Consider the following two vectors and a scalar:

$$\mathbf{p} = \begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2, \quad \mathbf{w} = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2, \quad \lambda = 2.$$

Then:

$$\mathbf{p} + \mathbf{w} = \begin{pmatrix} 1 + 4 \\ 4 + 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}, \quad \lambda \mathbf{p} = 2 \begin{pmatrix} 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 2 \\ 8 \end{pmatrix}.$$

Properties of vector addition and scalar multiplication (1/2)

$$\mathbf{p} + \mathbf{w} = \mathbf{w} + \mathbf{p}, \quad \forall \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (4)$$

$$(\mathbf{p} + \mathbf{w}) + \mathbf{u} = \mathbf{p} + (\mathbf{w} + \mathbf{u}), \quad \forall \mathbf{p}, \mathbf{w}, \mathbf{u} \in \mathbb{R}^n \quad (5)$$

$$\lambda(\mathbf{p} + \mathbf{w}) = \lambda\mathbf{p} + \lambda\mathbf{w}, \quad \forall \lambda \in \mathbb{R}, \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (6)$$

Properties of vector addition and scalar multiplication (2/2)

$$(\lambda + \mu)\mathbf{p} = \lambda\mathbf{p} + \mu\mathbf{p}, \quad \forall \lambda, \mu \in \mathbb{R}, \mathbf{p} \in \mathbb{R}^n \quad (7)$$

$$\lambda(\mu\mathbf{p}) = (\lambda\mu)\mathbf{p}, \quad \forall \lambda, \mu \in \mathbb{R}, \mathbf{p} \in \mathbb{R}^n \quad (8)$$

$$1 \cdot \mathbf{p} = \mathbf{p}, \quad \forall \mathbf{p} \in \mathbb{R}^n. \quad (9)$$

Summary

Operations:

- ▶ **Sum:** $\mathbf{p} + \mathbf{w}$ is obtained by adding the entries component by component
- ▶ **Scalar multiplication:** $\lambda \mathbf{p}$ is obtained by multiplying every entry by λ

Properties:

- ▶ The sum is commutative and associative
- ▶ Scalar multiplication is distributive with respect to the sum of vectors and to the sum of scalars
- ▶ $\lambda(\mu \mathbf{p}) = (\lambda\mu) \mathbf{p}$ and $1 \cdot \mathbf{p} = \mathbf{p}$

Scalar Product

Introduction: Scalar Product

- ▶ The scalar product (or dot product) is a fundamental operation that maps two vectors to a real number
- ▶ It is the basic ingredient of the matrix product and of the ℓ_2 norm
- ▶ We define it and list its main properties

Scalar product

Given two column vectors

$$\underbrace{\begin{pmatrix} p_1 \\ p_2 \\ \vdots \\ p_n \end{pmatrix}}_{\mathbf{p}} \in \mathbb{R}^n \quad \text{and} \quad \underbrace{\begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{pmatrix}}_{\mathbf{w}} \in \mathbb{R}^n$$

the expression

$$\mathbf{p}' \mathbf{w} = \sum_{j=1}^n p_j w_j \quad (10)$$

is called the **scalar product** of $\mathbf{p}$ and $\mathbf{w}$.

Example

Consider the two vectors $\begin{pmatrix} 1 \\ 4 \end{pmatrix} \in \mathbb{R}^2$ and $\begin{pmatrix} 4 \\ 1 \end{pmatrix} \in \mathbb{R}^2$. Their scalar product is:

$$(1 \quad 4) \begin{pmatrix} 4 \\ 1 \end{pmatrix} = 1 \cdot 4 + 4 \cdot 1 = 4 + 4 = 8.$$

Properties of scalar product (1/2)

$$\mathbf{p}' \mathbf{w} = \mathbf{w}' \mathbf{p}, \quad \forall \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (11)$$

$$\mathbf{p}' (\mathbf{w} + \mathbf{u}) = \mathbf{p}' \mathbf{w} + \mathbf{p}' \mathbf{u}, \quad \forall \mathbf{p}, \mathbf{w}, \mathbf{u} \in \mathbb{R}^n \quad (12)$$

$$\lambda (\mathbf{p}' \mathbf{w}) = (\lambda \mathbf{p})' \mathbf{w}, \quad \forall \lambda \in \mathbb{R}, \mathbf{p}, \mathbf{w} \in \mathbb{R}^n \quad (13)$$

Properties of scalar product (2/2)

From the definition we have:

$$\mathbf{p}' \mathbf{p} \geq 0, \quad \forall \mathbf{p} \in \mathbb{R}^n \quad (14)$$

$$\mathbf{p}' \mathbf{p} = 0 \iff \mathbf{p} = \mathbf{0} \quad (15)$$

Summary

Scalar product:

- ▶ $\mathbf{p}'\mathbf{w} = \sum_{j=1}^n p_j w_j$ is a real number (sum of the products of the components)
- ▶ It is commutative: $\mathbf{p}'\mathbf{w} = \mathbf{w}'\mathbf{p}$
- ▶ It is distributive: $\mathbf{p}'(\mathbf{w} + \mathbf{u}) = \mathbf{p}'\mathbf{w} + \mathbf{p}'\mathbf{u}$
- ▶ Scalars can be moved: $\lambda(\mathbf{p}'\mathbf{w}) = (\lambda\mathbf{p})'\mathbf{w}$
- ▶ $\mathbf{p}'\mathbf{p} \geq 0$, with equality if and only if $\mathbf{p} = 0$

Linear Combinations and Linear Independence



Introduction: Linear Combinations and Linear Independence

- ▶ Combining sums and scalar multiplications we obtain **linear combinations** of vectors
- ▶ We will learn to decide whether some vectors are **linearly independent**, i.e., whether none of them is redundant
- ▶ Finally, we introduce the **span** and the **basis** of $\mathbb{R}^n$: a basis allows us to describe every vector by its coordinates
- ▶ These concepts are needed for the rank of a matrix and for systems of linear equations

Definition 3: Linear combination

Given k vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ and k scalars $\lambda_1, \lambda_2, \dots, \lambda_k \in \mathbb{R}$, the vector

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k = \sum_{i=1}^k \lambda_i \mathbf{v}_i \in \mathbb{R}^n \quad (16)$$

is called a **linear combination** of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ with **coefficients** $\lambda_1, \lambda_2, \dots, \lambda_k$.

Zero vector and trivial linear combination

- ▶ We denote by $0 \in \mathbb{R}^n$ the **zero vector**, i.e., the vector whose entries are all equal to 0.
- ▶ Choosing $\lambda_1 = \lambda_2 = \dots = \lambda_k = 0$ we always obtain 0: this is called the **trivial** linear combination.

Example

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \in \mathbb{R}^3,$$

and the coefficients $\lambda_1 = 2$ and $\lambda_2 = -3$. The corresponding linear combination is

$$2\mathbf{v}_1 - 3\mathbf{v}_2 = \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ -3 \\ 3 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}.$$

Definition 4: Linearly independent vectors

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are **linearly independent** if the only linear combination equal to the zero vector is the trivial one, i.e.,

$$\lambda_1 \mathbf{v}_1 + \dots + \lambda_k \mathbf{v}_k = \mathbf{0} \implies \lambda_1 = \dots = \lambda_k = 0.$$

Otherwise, i.e., if there exist coefficients $\lambda_1, \lambda_2, \dots, \lambda_k$, **not all zero**, such that $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k = \mathbf{0}$, the vectors are **linearly dependent**.

Example

Consider the vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix} \in \mathbb{R}^2$.

The condition $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 = \mathbf{0}$ reads

$$\begin{cases} \lambda_1 + 3\lambda_2 = 0 \\ 2\lambda_1 + \lambda_2 = 0 \end{cases}$$

From the first equation $\lambda_1 = -3\lambda_2$; substituting in the second one we get $-6\lambda_2 + \lambda_2 = -5\lambda_2 = 0$, hence $\lambda_2 = 0$ and $\lambda_1 = 0$.

Therefore, $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent.

Example

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 4 \\ 3 \end{pmatrix} \in \mathbb{R}^2.$$

With the coefficients $\lambda_1 = 1$, $\lambda_2 = 1$, $\lambda_3 = -1$ (not all zero) we get

$$\mathbf{v}_1 + \mathbf{v}_2 - \mathbf{v}_3 = \begin{pmatrix} 1 + 3 - 4 \\ 2 + 1 - 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Therefore, $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$ are linearly dependent.

Example

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \in \mathbb{R}^3.$$

The condition $\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3 = \mathbf{0}$ reads

$$\begin{cases} \lambda_1 + \lambda_3 = 0 \\ \lambda_2 + \lambda_3 = 0 \\ \lambda_1 + \lambda_2 = 0 \end{cases}$$

From the first two equations $\lambda_1 = -\lambda_3$ and $\lambda_2 = -\lambda_3$; substituting in the third one we get $-2\lambda_3 = 0$. Hence $\lambda_1 = \lambda_2 = \lambda_3 = 0$, and the three vectors are linearly independent.

Example

Consider the vectors

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix} \in \mathbb{R}^3.$$

With the coefficients $\lambda_1 = 1$, $\lambda_2 = -2$, $\lambda_3 = 1$ we get

$$\mathbf{v}_1 - 2\mathbf{v}_2 + \mathbf{v}_3 = \begin{pmatrix} 1 - 8 + 7 \\ 2 - 10 + 8 \\ 3 - 12 + 9 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore, the three vectors are linearly dependent. Equivalently, $\mathbf{v}_3 = 2\mathbf{v}_2 - \mathbf{v}_1$ is a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$.

We now state some useful properties of linearly dependent vectors.

Observation 1: Sets containing the zero vector

Any set of vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ containing the zero vector $\mathbf{0}$ is linearly dependent.

Proof

Assume, without loss of generality, that $\mathbf{v}_1 = \mathbf{0}$.

Choosing $\lambda_1 = 1$ and $\lambda_2 = \dots = \lambda_k = 0$ we get

$$1 \cdot \mathbf{0} + 0 \cdot \mathbf{v}_2 + \dots + 0 \cdot \mathbf{v}_k = \mathbf{0},$$

which is a linear combination equal to $\mathbf{0}$ with coefficients not all zero.



Observation 2: Characterization of linear dependence

Let $k \geq 2$. The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are linearly dependent if and only if at least one of them is a linear combination of the others.

Proof (1/2)

($\Rightarrow$) If the vectors are linearly dependent, there exist coefficients, not all zero, such that

$$\sum_{i=1}^k \lambda_i \mathbf{v}_i = 0.$$

Let j be an index with $\lambda_j \neq 0$. Then

$$\mathbf{v}_j = \sum_{i \neq j} \left(-\frac{\lambda_i}{\lambda_j} \right) \mathbf{v}_i,$$

i.e., $\mathbf{v}_j$ is a linear combination of the others.

Proof (2/2)

($\Leftarrow$) If $\mathbf{v}_j = \sum_{i \neq j} \mu_i \mathbf{v}_i$ for some scalars μ_i , then

$$\sum_{i \neq j} \mu_i \mathbf{v}_i + (-1) \mathbf{v}_j = 0,$$

which is a linear combination equal to 0 whose coefficient of $\mathbf{v}_j$ is $-1 \neq 0$.



Observation 3: More than n vectors of $\mathbb{R}^n$

Any $k > n$ vectors of $\mathbb{R}^n$ are linearly dependent.

- ▶ We state this result without proof.
- ▶ For instance, any three vectors of $\mathbb{R}^2$ are linearly dependent (see the example with $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3 \in \mathbb{R}^2$ above).

Span and basis

- ▶ We now collect all the linear combinations of given vectors in a single set: their **span**
- ▶ Linearly independent vectors whose span is the whole $\mathbb{R}^n$ form a **basis**
- ▶ With respect to a basis, every vector is identified by a unique list of **coordinates**

Definition 5: Span

The **span** of the vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ is the set of all their linear combinations:

$$\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k) = \left\{ \lambda_1 \mathbf{v}_1 + \dots + \lambda_k \mathbf{v}_k : \lambda_1, \dots, \lambda_k \in \mathbb{R} \right\}.$$

Definition 6: Basis of $\mathbb{R}^n$

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k \in \mathbb{R}^n$ form a **basis** of $\mathbb{R}^n$ if

- ▶ they are linearly independent, and
- ▶ $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k) = \mathbb{R}^n$, i.e., every vector of $\mathbb{R}^n$ is a linear combination of them.

Observation 4: Coordinates with respect to a basis

*If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$ form a basis of $\mathbb{R}^n$, then every vector $\mathbf{x} \in \mathbb{R}^n$ can be written in a **unique** way as $\mathbf{x} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \dots + \lambda_k \mathbf{v}_k$. The coefficients $\lambda_1, \lambda_2, \dots, \lambda_k$ are called the **coordinates** of $\mathbf{x}$ with respect to the basis.*

Proof

Such coefficients exist because the span of the basis is $\mathbb{R}^n$.

If $\mathbf{x} = \sum_{i=1}^k \lambda_i \mathbf{v}_i = \sum_{i=1}^k \mu_i \mathbf{v}_i$, subtracting the two expressions we get

$$\sum_{i=1}^k (\lambda_i - \mu_i) \mathbf{v}_i = 0.$$

Since the vectors are linearly independent,

$$\lambda_i - \mu_i = 0, \quad \forall i \in \{1, 2, \dots, k\}.$$



Definition 7: Canonical basis

For $i \in \{1, 2, \dots, n\}$, we denote by $\mathbf{e}_i \in \mathbb{R}^n$ the vector whose i -th entry is equal to 1 and all other entries are equal to 0:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

The vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ form the **canonical basis** of $\mathbb{R}^n$.

Observation 5: The canonical basis is a basis

The vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ form a basis of $\mathbb{R}^n$, and every $\mathbf{x} \in \mathbb{R}^n$ satisfies

$$\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + \cdots + x_n \mathbf{e}_n. \quad (17)$$

- It can be shown that every basis of $\mathbb{R}^n$ consists of exactly n vectors.

Proof

For any scalars $\lambda_1, \dots, \lambda_n$ we have

$$\lambda_1 \mathbf{e}_1 + \dots + \lambda_n \mathbf{e}_n = (\lambda_1 \quad \lambda_2 \quad \dots \quad \lambda_n)'.$$

- ▶ This vector is equal to 0 only if $\lambda_1 = \dots = \lambda_n = 0$, hence the vectors are linearly independent.
- ▶ Choosing $\lambda_i = x_i$ we obtain (17), hence their span is $\mathbb{R}^n$.



Example

The vector $\mathbf{x} = (2 \ -5 \ 3)' \in \mathbb{R}^3$ can be written as

$$\mathbf{x} = 2 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} - 5 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + 3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = 2\mathbf{e}_1 - 5\mathbf{e}_2 + 3\mathbf{e}_3.$$

Example (1/2)

Consider the vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$.

- ▶ They are linearly independent: $\lambda_1 + \lambda_2 = 0$ and $\lambda_1 - \lambda_2 = 0$ give $\lambda_1 = \lambda_2 = 0$.
- ▶ Their span is $\mathbb{R}^2$: for every $(a \ b)' \in \mathbb{R}^2$,

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{a+b}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{a-b}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Hence $\mathbf{v}_1, \mathbf{v}_2$ form a basis of $\mathbb{R}^2$.

Example (2/2)

For instance, the coordinates of $\mathbf{x} = \begin{pmatrix} 3 & 1 \end{pmatrix}'$ with respect to the basis $\mathbf{v}_1, \mathbf{v}_2$ are

$$\lambda_1 = \frac{3+1}{2} = 2 \quad \text{and} \quad \lambda_2 = \frac{3-1}{2} = 1,$$

i.e.,

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Summary (1/2)

Linear combinations:

- ▶ $\sum_{i=1}^k \lambda_i \mathbf{v}_i$; the trivial one (all $\lambda_i = 0$) always gives 0

Linear independence:

- ▶ $\mathbf{v}_1, \dots, \mathbf{v}_k$ are linearly independent if only the trivial combination gives 0
- ▶ To check it: solve $\lambda_1 \mathbf{v}_1 + \dots + \lambda_k \mathbf{v}_k = 0$ for the λ_i
- ▶ A set containing 0 is linearly dependent
- ▶ Dependent $\iff$ one vector is a linear combination of the others
- ▶ More than n vectors of $\mathbb{R}^n$ are always linearly dependent

Summary (2/2)

Span and basis:

- ▶ $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ is the set of all linear combinations of $\mathbf{v}_1, \dots, \mathbf{v}_k$
- ▶ A basis of $\mathbb{R}^n$ is a set of linearly independent vectors whose span is $\mathbb{R}^n$
- ▶ Every vector has unique coordinates with respect to a basis
- ▶ The canonical basis $\mathbf{e}_1, \dots, \mathbf{e}_n$: $\mathbf{x} = x_1\mathbf{e}_1 + \dots + x_n\mathbf{e}_n$
- ▶ Every basis of $\mathbb{R}^n$ has exactly n vectors

Key Takeaways

Summary (1/4)

Vector Definition:

- ▶ A vector is a column of n real numbers: $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^n$
- ▶ The transpose $\mathbf{a}'$ converts a column vector to a row vector

Summary (2/4)

Vector Operations:

► **Sum:** $\mathbf{p} + \mathbf{w} = \begin{pmatrix} p_1 + w_1 \\ \vdots \\ p_n + w_n \end{pmatrix}$ (component-wise addition)

► **Scalar multiplication:** $\lambda \mathbf{p} = \begin{pmatrix} \lambda p_1 \\ \vdots \\ \lambda p_n \end{pmatrix}$ (multiply each component)

► **Scalar product:** $\mathbf{p}'\mathbf{w} = \sum_{j=1}^n p_j w_j$ (sum of component products)

Summary (3/4)

Key Properties:

- ▶ Vector addition is commutative and associative
- ▶ Distributive properties hold for scalars and vectors
- ▶ Scalar product is commutative: $\mathbf{p}'\mathbf{w} = \mathbf{w}'\mathbf{p}$
- ▶ Scalar product is distributive
- ▶ $\mathbf{p}'\mathbf{p} \geq 0$ with equality only for $\mathbf{p} = 0$

Summary (4/4)

Linear Combinations, Independence and Bases:

- ▶ A linear combination is $\lambda_1 \mathbf{v}_1 + \cdots + \lambda_k \mathbf{v}_k$
- ▶ Linearly independent: only the trivial combination gives 0
- ▶ The span is the set of all linear combinations
- ▶ A basis of $\mathbb{R}^n$ is linearly independent and spans $\mathbb{R}^n$; it has exactly n vectors and gives unique coordinates (e.g., the canonical basis $\mathbf{e}_1, \dots, \mathbf{e}_n$)

